Review: Identifying similar triangles

1

a

i A and C

ii All sides are in the same ratio.

b

i B and D

ii All corresponding angles are congruent.

2 A and B

3 Another pair of congruent angles must be added.

4

a No **b** Yes **c** No **d** No

5

a No **b** No

6

a The triangles are similar by Angle-angle similarity (AA∼).

b The triangles are not similar. The sides are not in the same proportion.

c The triangles are similar by Side-angle-side similarity (SAS∼).

d The triangles are similar by Side-side-side similarity (SSS∼).

e The triangles are similar by Side-angle-side similarity (SAS∼).

f The triangles are similar by Angle-angle similarity (AA∼).

g The triangles are similar by Side-angle-side similarity (SAS∼).

h The triangles are not similar. The sides are not in the same proportion.

7 Each of the angles measures $60\degree$. This means two equilateral triangles will always have three pairs of congruent angles, and so they will satisfy Angle-angle-angle (AAA) similarity condition.

Additional questions

8

a

i A and D ii HL

i B and D ii AA

c

i A and B ii HL

d

i C and D ii SSS

e

i A and C ii SAS

f

i B and C ii SAS

g

i B and D ii HL

h

i B and D ii AA

9

a $\triangle ABC$ and $\triangle ADE$ are similar by AA

b $\triangle JMN$ and $\triangle JKL$ are similar by AA

10 AA

11 $\frac{AB}{BD} = \frac{AC}{DE}$

12 Example answers:

- $\frac{FX}{GX} = \frac{3}{7}$
- EF and GH are parallel.

13

a The triangles don't have matching sides in the same ratio.

b $YZ = 9$